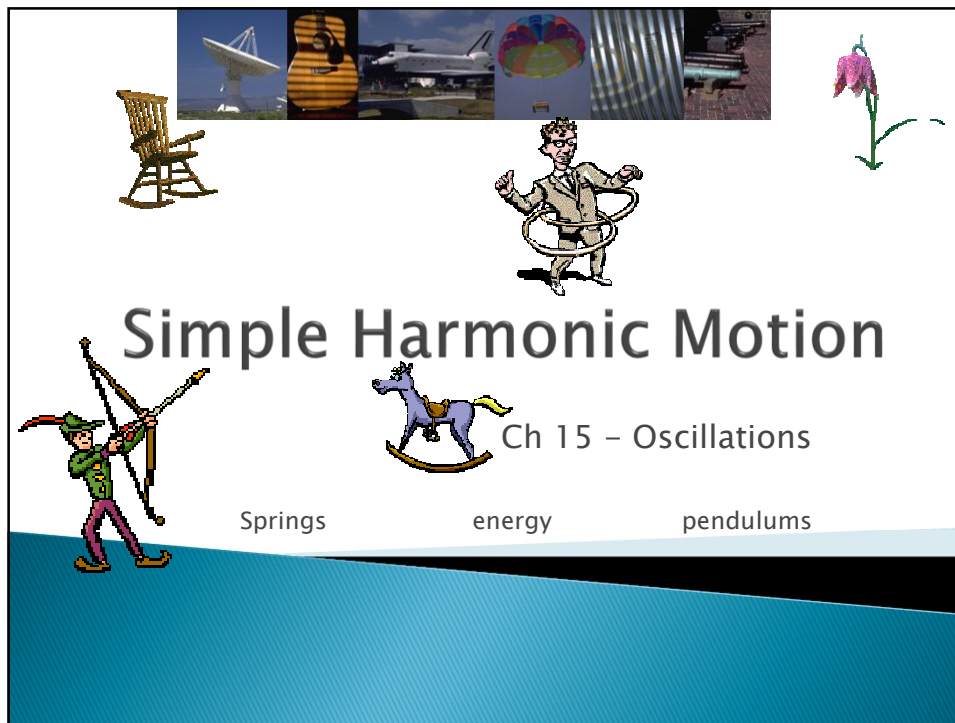




Simple Harmonic Motion (SHM)

- ▶ **Periodic motion or harmonic motion**
 - motion that repeats itself at regular intervals.
 - Particle/object moving back and forth repeatedly.
- ▶ **Frequency f**
 - number of completed oscillations per second.
 - SI unit: 1 hertz (Hz) = 1 oscillation/s
= 1 (s⁻¹)
- ▶ **Period T**
 - time for one complete oscillation.
 - $$T = \frac{1}{f}$$



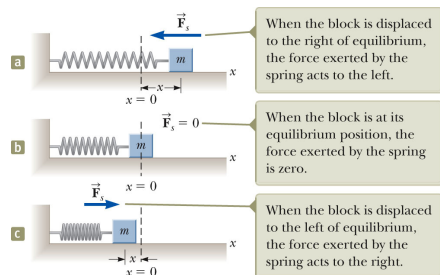
Motion of mass–spring system

- ▶ Block of mass, m , attached to the end of a spring.
- ▶ Block moves back and forth on a frictionless horizontal surface.
- ▶ For the spring we have a restoring force: $F = -kx$
- ▶ But also : $\sum F_x = ma_x$

▶ So
$$a_x = -\frac{k}{m}x$$

$$a \propto -x$$

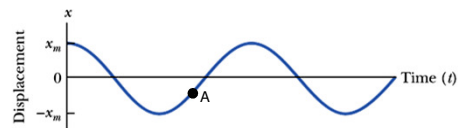
- ▶ Systems with this behaviour are said to exhibit **SIMPLE HARMONIC MOTION**



Problem

Consider the graphical representation of SHM as described mathematically by the equation we have just looked at. When the particle is at point A, what can you say about the particle's position and velocity?

- a) The position and velocity are both positive.
- b) The position and velocity are both negative.
- c) The position is positive and velocity is zero.
- d) The position is negative and velocity is zero.
- e) The position is negative and velocity is positive.
- f) The position is positive and velocity is negative.



7

Simple Harmonic Motion (SHM)

$$x(t) = A \cos(\omega t + \phi)$$

- ▶ Motion repeats after one period ($t + T$) – this is when the phase increases by 2π radians.

$$[\omega(t + T) + \phi] - (\omega t + \phi) = 2\pi$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

Period

- ▶ $f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$

Frequency

$$\omega = 2\pi f$$

8

Simple Harmonic Motion (SHM)

- Velocity of SHM**

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}[A \cos(\omega t + \phi)]$$

$$v(t) = -\omega A \sin(\omega t + \phi)$$

$$\text{Velocity amplitude: } v_{\max} = \omega A = \sqrt{\frac{k}{m}} A$$

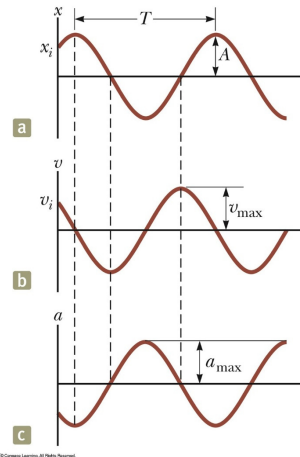
- Acceleration of SHM**

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}[-\omega A \sin(\omega t + \phi)]$$

$$a(t) = -\omega^2 A \cos(\omega t + \phi) = -\omega^2 x$$

$$\text{Acceleration amplitude: } a_{\max} = \omega^2 A = \frac{k}{m} A$$

In SHM the $a \propto x$ but opposite in sign, and both are related by ω^2 .



9

Example

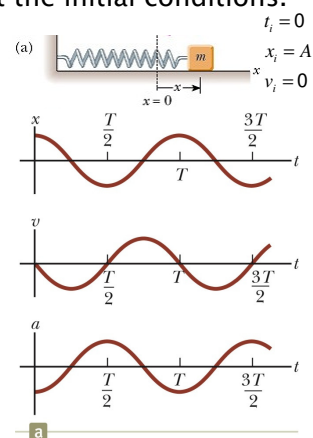
- A block is set into motion by pulling it from equilibrium by a distance A and releasing it from rest at $t = 0$. The solutions for the equations for $x(t)$ and $v(t)$ must reflect the initial conditions.

$$x(t) = A \cos(\omega t + \phi)$$

$$x(0) = A \cos(\omega(0) + \phi) = A$$

$$v(0) = -\omega A \sin(\omega(0) + \phi) = 0$$

This is satisfied when $\cos(\omega t + \phi) = 1$, therefore $\phi = 0$.



10

Example 2

- Consider the same system as previous, but now it is oscillating, and we define $t = 0$ as the block passes through the unstretched position while moving to the right.

$$x(t) = A \cos(\omega t + \phi)$$

$$x(0) = A \cos(\omega(0) + \phi) = 0$$

$$v(0) = -\omega A \sin(\omega(0) + \phi) = v_i$$

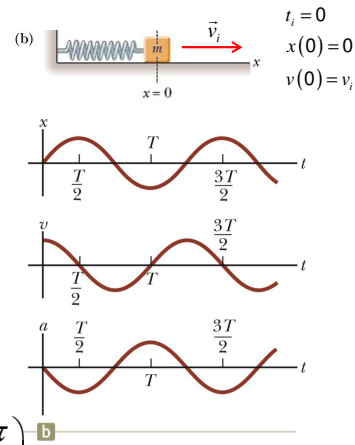
This is satisfied when $\cos(\omega t + \phi) = 0$, therefore $\phi = \pm \frac{\pi}{2}$.

The second condition the velocity and amplitude must be positive.

$$A = \mp \frac{v_i}{\omega}$$

$$\therefore \phi = -\frac{\pi}{2}$$

$$x(t) = \frac{v_i}{\omega} \cos\left(\omega t - \frac{\pi}{2}\right)$$



11

Problem

A cork floating on water moves in a simple harmonic manner. The peak-to-peak distance is 4 cm. $T = 1.0$ s. At $t = 0$ the cork is at its minimum height. What are the height and velocity after 10.5 periods?

12

Energy in simple harmonic motion

- ▶ Potential energy of a linear oscillator (spring) depends on how far the spring is stretched $x(t)$.

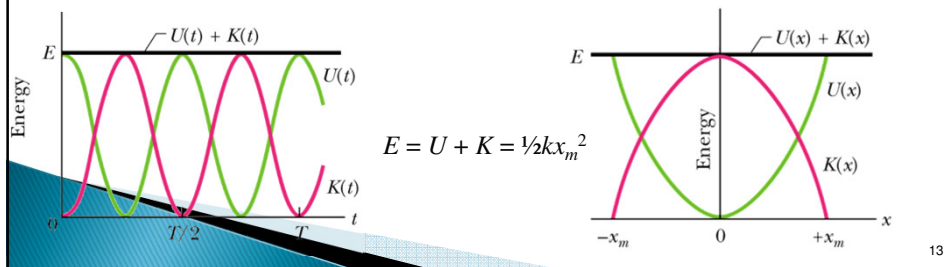
$$U(t) = \frac{1}{2} kx^2 = \frac{1}{2} kx_m^2 \cos^2(\omega t + \phi)$$

- ▶ Kinetic energy of the system is associated entirely with the mass and depends on its speed $v(t)$.

$$K(t) = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 x_m^2 \sin^2(\omega t + \phi)$$

$$K(t) = \frac{1}{2} k x_m^2 \sin^2(\omega t + \phi)$$

- ▶ Mechanical energy of oscillator - constant = $U + K$



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Energy in simple harmonic motion

- ▶ Mechanical energy of oscillator - constant = $U + K$

$$E = \frac{1}{2} k A^2 = U + K$$

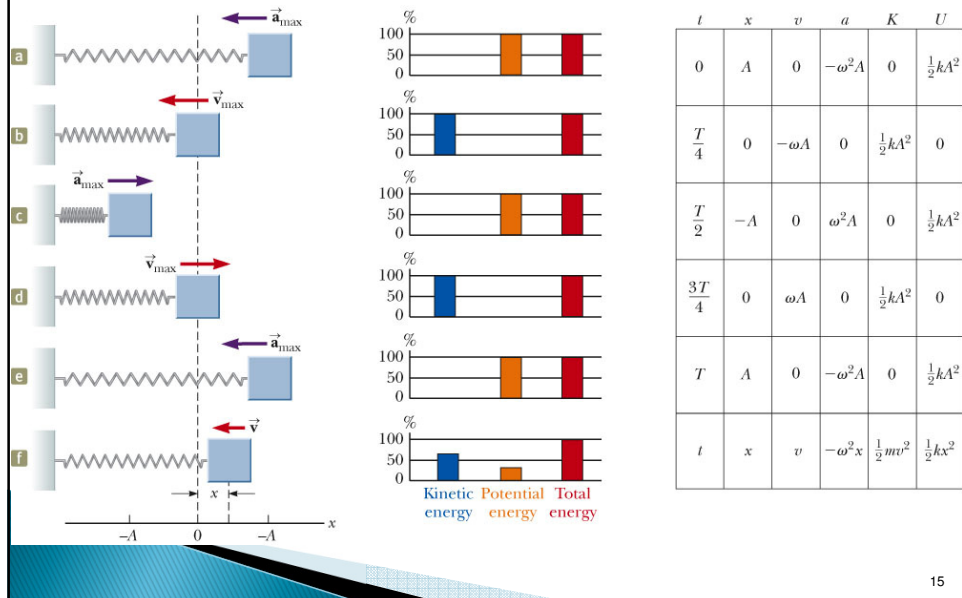
$$= \frac{1}{2} m v^2 + \frac{1}{2} k x^2$$

$$v = \sqrt{\frac{k}{m} (A^2 - x^2)} = \pm \omega \sqrt{A^2 - x^2}$$

Speed at any position x

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Energy in simple harmonic motion

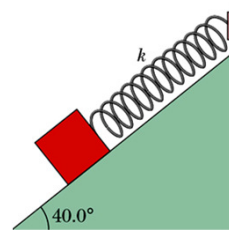


Problem

A body oscillates with simple harmonic motion according to the equation $x = (6.0 \text{ m})\cos[(3\pi \text{ rad/s})t + \pi/3 \text{ rad}]$. At $t = 2.0 \text{ s}$, what are the (a) displacement, (b) velocity, (c) acceleration, and (d) the phase of motion? Also, what are the (e) frequency and (f) the period of the motion?

Problem

In the figure a block weighing 14.0 N, which can slide without friction on an incline at an angle $\theta = 40^\circ$, is connected to the top of the incline by a massless spring of unstretched length 0.450 m and spring constant 120 N/m.

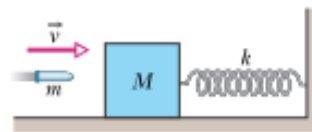


- How far from the top of the incline is the block's equilibrium position?
- If the block is pulled slightly down the incline and released, what is the period of the resulting oscillations?

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Problem

A block of mass $M = 5.4$ kg, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant $k = 6000$ N/m. A bullet of mass $m = 9.5$ g and velocity of magnitude 630 m/s strikes and is embedded in the block – see Figure. Assuming the compression of the spring is negligible until the bullet is embedded, determine:



- the speed of the block immediately after the collision and
- the amplitude of the resulting simple harmonic motion.
- What fraction of the original kinetic energy of the bullet appears as mechanical energy in the harmonic oscillator?

18

SHM & circular motion

- ▶ Read Section 15–4

19

Pendulums – Simple Pendulum

- ▶ **Simple pendulum:** mass m suspended from a light unstretchable, massless string of length L .

- ▶ Forces on mass:

- weight mg and
- tension T .

- ▶ tangential component: $mg \sin \theta$
radial component: $mg \cos \theta$

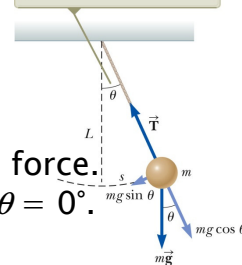
- ▶ Tangential component produces restoring force.

Brings mass back to equilibrium position $\theta = 0^\circ$.

$$F_t = ma_t \quad \rightarrow \quad -mg \sin \theta = m \frac{d^2 s}{dt^2}$$

- ▶ Tangential acceleration to position s

When θ is small, a simple pendulum's motion can be modeled as simple harmonic motion about the equilibrium position $\theta = 0$.



20

Pendulums – Simple Pendulum

- ▶ $s = L\theta$ and L is constant

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta$$

- ▶ Compare to $\frac{d^2x}{dt^2} = -\frac{k}{m}x$, the criteria for SHM.

- ▶ If we assume small angles: $\theta < 10^\circ$

- ▶ Then $\sin\theta \approx \theta$ (See Table 15.1)

- ▶ and $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$

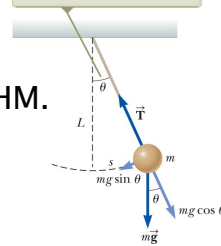
$$\theta = \theta_{\max} \cos(\omega t + \phi)$$

- ▶ Then

$$\omega = \sqrt{g/L}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{L/g}$$

When θ is small, a simple pendulum's motion can be modeled as simple harmonic motion about the equilibrium position $\theta = 0$.



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Pendulums – Simple Pendulum

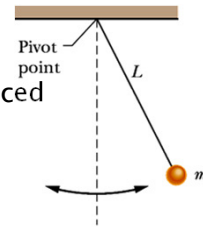
- ▶ $T = 2\pi\sqrt{\frac{L}{g}}$

- ▶ Period is ONLY dependent on the length of the string and g .
- ▶ Simple pendulum used in clocks.

22

Problem

A pendulum has a length of 2 m. The mass is displaced from the vertical position by giving it a small push. The maximum speed is 0.11 m/s. What is the maximum horizontal displacement?



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Pendulums – Physical Pendulum

- ▶ If an object oscillates about a fixed axis not passing through COM – cannot treat it as a simple pendulum – it is a Physical Pendulum

- ▶ Consider: Rigid object rotating about O .
- ▶ Gravitational force $\rightarrow$ torque $\tau = -mgd \sin \theta$

$$\sum \tau_{net} = I\alpha$$

$$I \frac{d^2 \theta}{dt^2} = -mgd \sin \theta$$

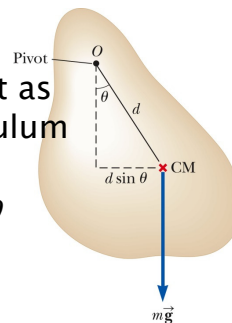
- ▶ For small angles: $\frac{d^2 \theta}{dt^2} = -\frac{mgd}{I} \theta = -\omega^2 \theta$

- ▶ So $\theta = \theta_{max} \cos(\omega t + \phi)$

- ▶ This results in:

$$\omega = \sqrt{mgd/I}$$

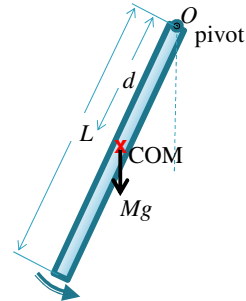
$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{mgd}}$$



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Example 15.6

A uniform rod of mass M and length L is pivoted about one end and oscillates in a vertical plane. Find the period of oscillation if the amplitude of the motion is small.

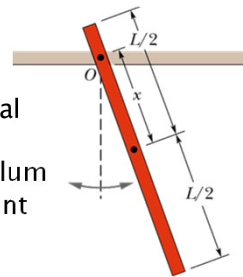


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Problem

A stick of length $L = 1.00$ m oscillates as a physical pendulum.

- Derive an equation for the period of the pendulum in terms of L and x the distance from the pivot point to the centre of mass.
- For what value of x/L is the period a minimum?
- What is this minimum period?



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Problems

- ▶ Do all Sample Problems from the relevant sections